

Computation of function of multivector and matrix using generalised spectral basis

Arturas Acus

Institute of Theoretical Physics and Astronomy, Vilnius University, Sauletekio 3, LT-10257 Vilnius

Recursive formulas for explicit computation of generalised spectral basis are presented. The generated spectral basis allows straightforward computation of functions of matrices and multivectors of arbitrary Clifford geometric algebra $Cl_{p,q}$. The formulas are based on analysis of the roots of minimal polynomial and covers defective matrices/multivectors case. The method avoids matrix/multivector inversion and, therefore, often provides simplest symbolic answers. It can find application in Clifford algebra analysis, computation of Krylov spaces, etc.